



SHOPEZ PLATFORM SUITE

PROJECT DOCUMENTATION

Functional Specification Document (FSD)

System Specifications, Features & Functional Flowcharts

PROJECT NAME: ShopEZ Premium E-Commerce Platform

PREPARED BY: P Pranay

DATE: June 2026

VERSION / STATUS: v1.0.0 / Final Approved

1. Project Overview & Scope

The ShopEZ platform is a premium full-stack MERN e-commerce application designed to emulate state-of-the-art consumer shopping marketplaces like Amazon and Flipkart. The application focuses on delivering rich customer experience modules including interactive search autocomplete, responsive visual category mappings, robust multi-step checkout sequences, coupon promotions, user wallets, and direct seller/administrative content controls.

1.1 System Actors & Permissions

- Guest Visitors: Browse products, search the catalog, filter by attributes, and view product details/FAQ lists.
- Registered Customers: Manage personal profiles, select/store multiple delivery addresses, add items to cart and wishlist, apply valid coupon codes, execute payments using simulated wallets or UPI, and track order timeline status.
- System Administrators: Full catalog management (CRUD products), order status progression updates, customer status management (toggle block/unblock), coupon management, and analytics tracking.

2. Detailed System Features

ShopEZ includes several premium modules designed for visual excellence and optimal operational throughput:

- Premium UI Aesthetics: Utilizes modern CSS structures, smooth hover animations, and skeleton loaders to bridge the gap during database transactions.
- Autocompleting Search Bar: Suggests matching products in real-time as users type, keeping track of local user search histories.
- Multi-Step Checkout: Separates address confirmation, order overview with coupon application, and payment simulations (Wallet, UPI, Credit Card, COD).
- Administrative Controls: Built-in Chart.js analytics dashboard tracking monthly revenue, category inventory distributions, and active coupon triggers.

SYSTEM FUNCTIONAL FLOWCHART DIAGRAM



